

Luis Carranza

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EDUCATION

- **PhD in Computer Science**, Universitat Politècnica de Catalunya Jan. 2024 - Present
 - Thesis: Training Physics-based Characters to Accomplish Complex Tasks such as Climbing.
 - Advisors: Oscar Argudo and Carlos Andujar.
 - Tools & Methods: Deep Reinforcement Learning, Isaacgym, IsaacLab.
- **MSc in Computer Science**, Universitat Politècnica de Catalunya Sep. 2021 - Jul. 2023
 - Thesis: Simulation and authoring of lava flows in virtual terrains.
 - Advisor: Oscar Argudo.
 - GPA: 8.55/10.0
- **BSc in Informatics Engineering**, Pontificia Universidad Católica del Perú Mar. 2015 – Aug. 2020
 - Thesis: 3D Reconstruction of Incomplete Archaeological Objects through Deep Learning using multi-view images.
 - Advisor: Ivan Sipiran.
 - GPA: 15.13/20.0

PUBLICATIONS

- **Luis Carranza**, Oscar Argudo, and Carlos Andujar. PHA: Part-wise Heterogeneous Agents with Reusable Policy Priors for Physics-Based Motion Synthesis. In Proc. ACM Comput. Graph. Interact. Tech. (SCA '25), 2025.
- Diego Mateos, **Luis Carranza**, Antoni Susin, and Oscar Argudo. Simulation of Mechanical Weathering for Modeling Rocky Terrains. In Spanish Computer Graphics Conference (CEIG '24), 2024.

WORK EXPERIENCE

- **Research Intern**, École de technologie supérieure ÉTS (Montreal, Canada) Feb. 2026 – Jun. 2026
 - Development of novel algorithms for motion synthesis of physics-based characters, with applications in real-time and interactive computer graphics.
- **Freelance**, Hyper Online Inc. (YCombinator Startup) (Remote) Feb. 2024 – Jan. 2025
 - Designed and implemented a pipeline for real-time face, body and hands motion capture from a webcam, supporting different 3D avatar formats.
- **Research Intern**, Universitat Politècnica de Catalunya (Barcelona, Spain) Oct. 2022 – Oct. 2023
 - Contributed to an HP (Hewlett-Packard) project developing a library in C++ to transform 3D models into slices for 3D printing.
- **Software Engineer**, Tuxpas (Meta Workplace & AWS Partner) (Remote) Jan. 2021 – Dec. 2021
- **Project Management Intern**, Huawei (Lima, Perú) Aug. 2020 – Dec. 2020
- **Full-Stack Developer Intern**, Ernst & Young (Lima, Perú) May 2018 – Oct. 2018

AWARDS

- **Becas Santander Scholarship**, Banco Santander España, 2025.
- **PhD FPI Grant**, Spanish Government, 2023.
- **Research Initiation Grants**, Universitat Politècnica de Catalunya, 2022.
- **Becas Santander Scholarship**, MIT Professional Education, 2020.
- **CINDA International Student Exchange Scholarship**, PUCP, 2019.

MISCELLANEOUS

- Teaching Assistant, Interaction and Interface Design, Spring and Fall 2025.
- Winner, Generative AI Agents Developer Contest by NVIDIA and LangChain, 2024.
- Winner, Huawei Challenge, HackUPC, 2022.
- Winner, DigiEduHack (Worldwide), European Commission, 2020.
- Winner, Facebook Challenge, HackUPC, 2019.